

Zhenchuan Yang (1997.12)

Institute of Space and Earth Information Science, the Chinese University of Hong Kong

Email: zhenchuan.yang@link.cuhk.edu.hk

Website: <https://zhenchuan yang.github.io/>

EDUCATION

Institute of Space and Earth Information Science, the Chinese University of Hong Kong (CUHK)

PhD, Earth System and GeoInformation Science

08/2022-03/2026

Supervisor: Professor Mei-Po Kwan

School of Management Science and Real Estate, Chongqing University (CQU)

M.S., Management Science and Engineering

09/2019-06/2022

Supervisor: Professor Liyin Shen

Awards: Outstanding Student

B.E., Construction Management

Awards: National Encouragement Scholarship, Outstanding Cadre, Outstanding Student

09/2015-06/2019

SELECTED PUBLICATIONS

- **YANG Zhenchuan**, KWAN Mei-Po, ZHANG Yan, The dynamic interplay between greenspace, air pollution, physical activity, and mental health across mobility-based geographic contexts. *Environmental Research*. (SCI Q1, IF 7.7, in peer review)
- **YANG Zhenchuan**, KWAN Mei-Po, The interplay between actual and perceived air pollution, physical activity, and mental health across mobility-based geographic contexts: A double machine learning approach. *Social Science & Medicine*. (SCI Q1, SSCI Q1, IF 5.0, in revision)
- **YANG Zhenchuan**, KWAN Mei-Po, ZHANG Yan. Reassessing and Optimizing the Treatment Effect of Greenspace on Air Pollution Conditioned on Seasons, Vegetation Types, and Street Structures. *Environmental Science & Technology*. (SCI Q1, IF 11.3, in revision)
- **YANG Zhenchuan**, Ren, Y., Shen, L., Liao, X., & Kwan, M.-P. (2025). Spatiotemporal evolutions and drivers of ground-level ozone in China (2015–2020): A GTWR-Kriging approach. *Environmental Research*, 279, 121748. (SCI Q1, IF 7.7, published)
- **YANG Zhenchuan**, Kwan, M.-P., Liu, D., & Huang, J. (2025). How objective and subjective greenspace, combined with air and noise pollution, impacts mental health through the mediation of physical activity. *Urban Forestry & Urban Greening*, 105, 128683. (SCI Q1, SSCI Q1, IF 6.7, published)
- **YANG Zhenchuan**, Huang J., Kwan M.-P., & Liu D. (2024). The interplay among individuals' distress, daily activities, and perceptions of COVID-19 and neighborhood cohesion: A study using network analysis. *PLOS ONE*, 19(1), e0293157. (SCI Q2, IF 2.6, published)
- Shen, L., **YANG, Zhenchuan***, Du, X., Wei, X., & Chen, X. (2022). A health risk-based threshold method to evaluate Urban Atmospheric Environment Carrying Capacity in Beijing-Tianjin-Hebei (BTH) Region. *Environmental Impact Assessment Review*, 92, 106692. (*corresponding author, SSCI Q1, IF 11.2, published)

SELECTED PROJECTS

- **Deploying Geospatial Big Data and Real-time Mobile Sensing to Assess the Health Impacts of Individual Exposure to Green/blue Spaces, Light at Night, Air Pollution, and Noise (GLAN)**
08/2022-Present, *Hong Kong Research Grants Council (RGC) Collaborative Research Fund, No. C4023-20GF*, Ongoing, as a key contributor
- **Assessment and policy study on resources and environmental carrying capacities in large-scale cities under the background of big data**
09/2019-06/2022, *Major National Social Science Fund Project, No. 17ZDA062*, Completed, as a key contributor
- **High-resolution spatial and temporal distribution simulation of atmospheric environmental carrying capacity and its influencing factors in Beijing-Tianjin-Hebei Region -- Based on GWR-LUR Model**
09/2019-06/2022, *Graduate Student Research Innovation Project, No.CYS20037*, Completed, as a leader
- **Life Cycle Cost Analysis of Construction Equipment with Integrated Environmental Effects**

PROFILE

Research interest: Causal inference and urban sustainable development under the intersection of GIS, data science, environmental science, human mobility, and public health.

Professional skills: Advanced geostatistics, machine learning algorithms, simulations, and manipulation of open-sourced big data and first-hand survey data, based on Python, R, ArcGIS Pro, QGIS, Stata, EViews, and SPSS.

Teaching experiences: Teaching assistant in courses on *Introduction to Environmental Information, Health and Public Management (ESGS 5011)*, *Health and Public Management*, and *Earth System Science (ESGS 5001)* at CUHK.